

WASP-1b

R-1175

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_191106 · created 2026-07-31T03:46:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458793.653 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1192 +/- 0.01
Transit depth (Rp/Rs)^2	1.42 +/- 0.24 [%]
Orbital Inclination (inc)	85.66 +/- 1.38
Airmass coefficient 1 (a1)	0.989 +/- 0.012
Airmass coefficient 2 (a2)	0.013 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.25 %
Best Comparison Star	#3 - [304, 166]
Optimal Aperture	3.46
Optimal Annulus	6.46
Transit Duration (day)	0.1365 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1192 ± 0.01 vs 0.1036 (deviation 1.04σ)
Duration fit vs published	0.1365 d vs 0.1567 d (deviation 1.73σ)
Mid-time O–C	2.2 min
Depth SNR	8.0
Residual scatter	1.25 %

Light curve

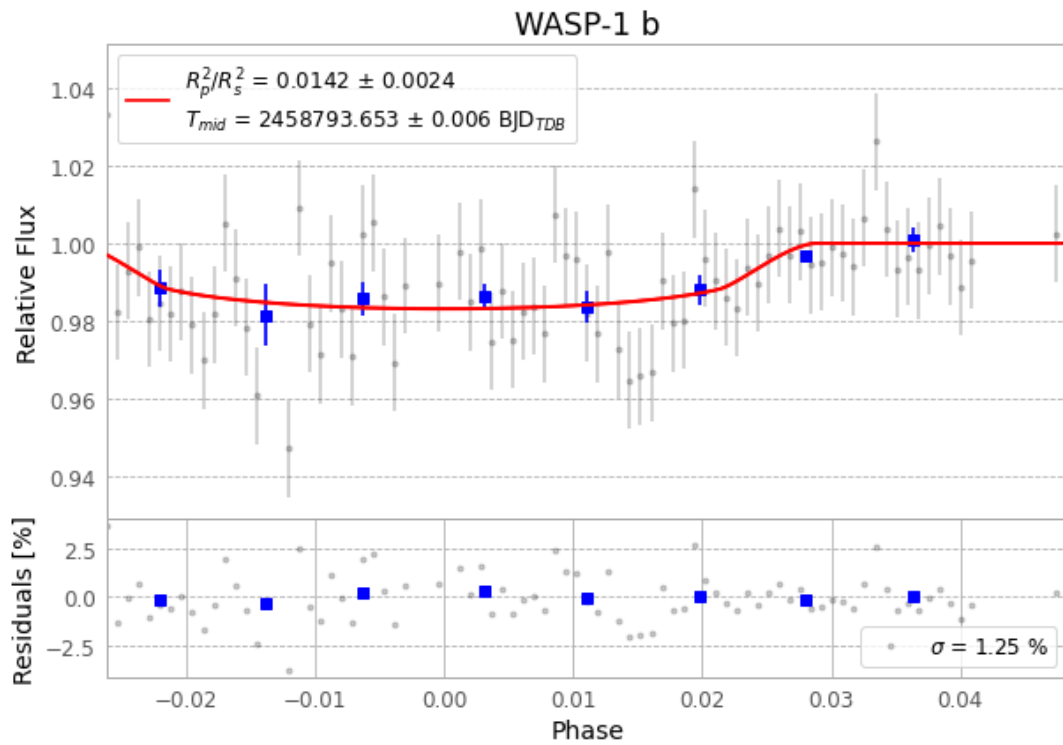


Figure 1 — FinalLightCurve_WASP-1 b_2019-11-05.png (SHA-256 2be58eff8052b98e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 02559c8b1de7ba16...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), WASP-1191106014812.FITS → WASP-1191106062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-191106.log (9d4e79d1779f...), FinalLightCurve_WASP-1 b_2019-11-05.png (2be58eff8052...), FinalLightCurve_WASP-1 b_2019-11-05.csv (ddaf4b88a23f...)

Compute resources

Wall time 235.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:46Z.